# Formatting Instructions for TMLR Journal Submissions

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## Abstract

Factual hallucinations in language models are often plausible substitutions: a model gives the wrong city, author, or entity of the right semantic type. We study a first-token version of this failure by separating *read* from *write*. Read measures whether the gold answer token is decodable from intermediate residual states, controlling for type priming with same-relation decoys. Write measures whether that token is top-ranked by the final readout.

Across models, many direct-generation failures are fact-specifically readable under a conservative hard-decoy criterion, yet the gold token remains below another candidate at the final readout. This dissociation is not explained by answer-token unembedding norms. Instead, first-token selection depends on the residual state's projection along the answer's unembedding direction. Calibrated interventions that set failed examples to successful-example levels of answer-direction support recover first-token selection in most models, while matched random perturbations do not. In some models, recovery requires changing both answer support and support for the selected alternative.

These results support a selection-margin account of one route to factual error. The model may carry decodable evidence for the correct answer, but the final readout can assign higher support to a plausible alternative, often a same-type or higher-frequency candidate. In this setting, hallucination is not simply absence of factual evidence, but a failure of that evidence to determine the selected first token.

## 1 Introduction

Large language models often fail on factual questions in a distinctively plausible way. Asked where a person was born, a model may answer with the wrong city; asked who wrote a book, it may produce another author. These errors are usually described as factual hallucinations: fluent, type-appropriate answers that are false. But this description leaves open a mechanical question. When the model gives the wrong answer, is the correct answer absent from the computation, or is it decodable internally without receiving enough final-readout support to begin the generated answer?

A natural way to approach this question is to compare generation with recognition. A model that fails to answer a question directly may still choose the correct answer from options, produce it under a paraphrased prompt, or assign it elevated probability under teacher forcing. Such cases are often described as a gap between what the model "knows" and what it says. We use this intuition as motivation, but find it unstable as a measurement target: reasonable recognition-like probes disagree about which facts count as available. If "known but not generated" depends strongly on the probe, then a mechanism for that gap inherits the ambiguity of the probe.

We therefore replace the epistemic question with a mechanical one. On a single forward pass, we distinguish *read* from *write*. Read measures whether the first content token of the gold answer is decodable from intermediate residual states before the final readout. Write measures whether that token receives enough final-readout support to be top-ranked in the next-token distribution. This lets us study a narrower and more reproducible phenomenon: cases where fact-specific evidence for the correct answer token is decodable inside the computation, yet the token is not selected.

The key challenge is to separate fact-specific readability from type priming. A birthplace prompt may activate city names in general, even if the model has not recovered the correct city. We therefore compare the gold answer with type-matched decoys drawn from the same relation pool. An answer counts as readable only when its intermediate decodability exceeds that of same-type alternatives. We also compare intermediate read with final-layer read: final-layer read nearly restates the output distribution, whereas intermediate read measures answer evidence before final readout selection has fully resolved.

Across models, we find a large read/write dissociation. Many direct-generation failures have positive type-controlled priming margin, meaning that the gold answer token is more readable than same-relation decoys at intermediate depth. Under a stricter hard-decoy criterion, the gold token must exceed the strongest same-type decoy; even then, a substantial fraction of failed generations remain readable, and these readable tokens are usually not top-ranked at the final readout. Prefix controls show that the effect is not driven by ambiguous first-token prefixes, and sequence-level read measures show a weaker but related pattern at the answer-string level. Paired paraphrase controls further show that the same fact can be selected or not depending on prompt wording: successful paraphrases have stronger intermediate read, stronger answer-direction support, and better final selection margin than failed paraphrases of the same fact.

We then ask what controls whether a readable answer token is selected at the final readout. The answer logit decomposes into the residual state's projection along the answer's unembedding direction and the fixed norm of that unembedding vector. Empirically, successful and failed cases differ mainly in the projection term. We use calibrated interventions as counterfactual tests: if a failed item is assigned answer-direction support typical of successful generations, does the gold token become top-ranked? In most models the answer-up intervention recovers first-token selection far more often than matched random perturbations. In Qwen2.5-3B, however, answer-up alone is ineffective and recovery appears only when answer support is raised while support for the selected alternative is reduced. This heterogeneity aligns with the model's layerwise trajectory profile and supports a relative-support account rather than a uniform single-factor account.

Finally, we analyze which candidate is selected when the correct answer token is not. The selected alternative is often a plausible token of the same semantic type and is frequently more common than the gold answer. This connects the read/write dissociation to the geometry of the output map. We find that token frequency is represented primarily by a shared direction in the unembedding rather than by larger output-vector norms. Frequency-direction ablations change which token is selected and shift selection toward lower-frequency tokens, but rarely recover the gold answer token. Training and toy-model analyses suggest two separable effects: directional frequency structure appears early and remains stable, while norm-based frequency coupling can decay under weight decay with co-evolving representations.

Our contribution is therefore not a new binary test for whether a model knows a fact, nor a full account of multi-token answer generation. Instead, we give a selection-margin account of one route to factual error at the first answer token. A model may contain decodable, fact-specific evidence for the correct answer token, but if that evidence receives insufficient support at the final readout, generation can begin with a more strongly supported, frequent, same-type alternative. On this view, some hallucinations are not failures to enter the right semantic neighborhood. They are failures to select the correct instance within it.

## 2 Related Work

Factual hallucination is usually studied as an output-level failure: a model generates fluent text that is unsupported, unverifiable, or false. Benchmarks such as TruthfulQA test whether models reproduce common falsehoods rather than truthful answers (**?**), while factuality metrics such as FActScore evaluate long-form generations by decomposing them into atomic claims (**?**). Other work detects hallucinations through consistency across samples or queries, as in SelfCheckGPT and related black-box approaches (**??**). These methods characterize when generated text is unreliable. Our work asks a different question: in a factual generation failure, what happens to the correct answer token inside the model before the final readout selects another token?

A long line of work probes whether pretrained language models store factual associations in their parameters. LAMA framed masked language models as implicit knowledge bases queried through cloze prompts (**?**), and

PopQA showed that parametric recall is strongly shaped by entity popularity and long-tail frequency (?). Related studies connect factual QA accuracy to the amount of relevant pretraining evidence, showing that rare facts are much harder for models to memorize and retrieve (?). We use PopQA-style subject–relation–object facts, but our focus is not factual accuracy alone. Instead, we separate intermediate decodability of the gold answer token from its final readout rank, allowing us to study cases where answer evidence is detectable but not selected.

Several recent works study whether internal activations contain information that differs from what the model says. Eliciting latent knowledge methods search for truth-related directions or probes in activation space (?), and later work investigates hidden-state signals of hallucination risk or untruthful behavior (??). These approaches are close in motivation to ours: they distinguish internal availability from generated output. Our contribution is narrower and more token-mechanical. Rather than training a classifier for truth or hallucination, we define read as logit-lens decodability of the gold answer token under type-matched controls, and write as final-readout selection of that same token.

Mechanistic studies of factual recall have localized causal contributions to particular modules and layers. Work on knowledge neurons identifies units associated with factual associations (?), feed-forward layers have been interpreted as key-value memories (?), and causal tracing plus model editing methods such as ROME and MEMIT show that specific factual associations can be localized and modified in transformer activations or weights (??). Our work is complementary. We do not claim to localize the circuit storing a fact. Instead, we analyze a later readout phenomenon: even when the gold answer is decodable, final selection can assign greater support to another candidate.

Our read measure is related to logit-lens analyses, which decode intermediate residual states through the model's output map. The tuned lens refines this idea by learning layer-specific affine translators and shows that logit-lens predictions can be biased or brittle at intermediate depths (?). We therefore treat intermediate read as decodability under a specified probe, not as a literal claim that the model has fully resolved the answer. The hard-decoy controls, read-definition sweep, prefix checks, and sequence-level analyses are designed to make the read/write dissociation less dependent on a single optimistic intermediate decoding choice.

The final readout of an autoregressive language model maps a residual state into logits through the unembedding matrix. Prior work has shown that embedding and output spaces encode frequency-related structure: token frequency can distort contextual embedding geometry (?), embedding norms can encode information-theoretic quantities (?), and output embeddings can contain a common direction related to token probabilities or frequencies (?). Our results connect this geometry to factual errors. We find that frequency is represented primarily directionally rather than through unembedding row norms, and that ablating the frequency direction changes which token is selected while rarely recovering the gold answer. Thus frequency-linked geometry shapes final readout selection, but it does not by itself explain the read/write dissociation.

Taken together, prior work shows that models store factual associations, that internal states can reveal information not expressed in the output, and that the output map has structured geometry. We show that the correct answer can be fact-specifically decodable under type-controlled intermediate probes, yet remain below a plausible alternative at the final readout.

## 3 Final-readout geometry

We describe the read/write dissociation as a negative selection margin under the model's final readout. For item $i$, let $h_i = \mathrm{LN}_f(h_i^{(L)})$ be the normalized final residual-stream state at the answer position, and let $W_U[v]$ be the unembedding vector for vocabulary token $v$. The final logit assigned to token $v$ is

$$z_{i,v} = \langle h_i, W_U[v] \rangle. \tag{1}$$

Generation selects the top-ranked token under this distribution. For the gold answer token $a_i$, define the highest-ranked alternative candidate

$$b_i = \arg\max_{v \neq a_i} z_{i,v}, \tag{2}$$

and the answer's final selection margin

$$M_i(a_i) = z_{i,a_i} - z_{i,b_i}. \tag{3}$$

The gold answer is selected exactly when $M_i(a_i) > 0$. When $M_i(a_i) < 0$, some alternative candidate receives higher final-readout support than the gold answer.

This separates two quantities that are often conflated. The first is whether fact-specific evidence for the answer is decodable before the final readout. The second is whether the final residual state assigns the answer enough support to make it top-ranked. Our read measure addresses the first quantity; write addresses the second. A readable-but-unselected answer is therefore not paradoxical: it is an answer with measurable intermediate decodability but insufficient final selection margin.

The relevant final-readout quantity is the residual state's projection onto the answer's unembedding direction. Let

$$u_{a_i} = \frac{W_U[a_i]}{\|W_U[a_i]\|} \tag{4}$$

be the unit unembedding direction for the answer token. Then

$$z_{i,a_i} = \langle h_i, u_{a_i} \rangle \|W_U[a_i]\|. \tag{5}$$

The answer logit factors into a state-dependent projection term and a fixed unembedding-norm term. We call

$$\pi_i(a_i) = \langle h_i, u_{a_i} \rangle \tag{6}$$

the answer-direction support. If answer-token norms are similar across successful and failed cases, differences in selection must arise primarily from how strongly the final residual state points in the answer direction.

This explains why intermediate readability need not imply generation. At an intermediate layer $\ell$, the answer may be decodable under the logit lens:

$$\text{Read}_i^{\text{int}} = \max_{\ell/L \in [0.4, 0.9]} \log\text{softmax}\left( W_U \text{LN}_f(h_i^{(\ell)}) \right)_{a_i}. \tag{7}$$

But this quantity is not the final readout. Between the layer where the answer is most readable and the final layer, the model may assign greater final support to another candidate token. The generated token is determined by final rank, not by whether the answer has nonzero intermediate evidence.

The causal prediction follows directly. If insufficient answer-direction support is limiting selection, then increasing the residual projection along $u_{a_i}$ should make the gold answer more likely to be selected. We test this by replacing the answer-direction component of the final residual state while preserving the orthogonal component:

$$h_i' = h_i - \langle h_i, u_{a_i} \rangle u_{a_i} + m u_{a_i}. \tag{8}$$

This changes the answer logit by

$$\Delta z_{i,a_i} = (m - \pi_i(a_i)) \|W_U[a_i]\|, \tag{9}$$

while changing other logits only through their alignment with $u_{a_i}$. Recovery under answer-direction patching, together with matched random-direction controls, tests whether answer-direction support is a causal variable for selection.

The same geometry also clarifies why many factual errors are plausible. A prompt may activate a relation-specific region of the output space: city tokens for birthplace questions, author tokens for authorship questions, and so on. The gold answer must then be assigned more final support than plausible same-type alternatives. If a same-type alternative $b_i$ receives a higher final logit, the model selects a fluent but false answer even when the gold answer is internally decodable.

Frequency can further shape this final ranking. Suppose token frequency is partly represented by a shared unembedding direction $r$, so that

$$W_U[v] = \beta_v r + s_v, \tag{10}$$

where $\beta_v$ correlates with token log-frequency and $s_v$ is the remaining token-specific component. Then

$$z_{i,v} = \beta_v \langle h_i, r \rangle + \langle h_i, s_v \rangle. \tag{11}$$

For an alternative candidate $b_i$, the gold answer is not selected when

$$\underbrace{\langle h_i, s_{b_i} \rangle - \langle h_i, s_{a_i} \rangle}_{\text{candidate-specific support}} + \underbrace{(\beta_{b_i} - \beta_{a_i})\langle h_i, r \rangle}_{\text{frequency-linked support}} > 0. \tag{12}$$

Thus a frequent same-type alternative can receive higher final-readout support through both relation-level compatibility and frequency-linked geometry.

On this account, one route to factual hallucination is a failure of final selection. The model may contain decodable, fact-specific evidence for the correct answer, but generation depends on whether that evidence is expressed strongly enough at the final readout. When it is not, the selected token can be a plausible alternative candidate: semantically close to the expected answer type, but factually wrong.

# 4 Methodology

We evaluate frozen autoregressive transformer language models from the Llama, Qwen, and Mistral families, including both base and instruction-tuned variants. Factual prompts are drawn from PopQA subject–relation–object triples with answer aliases. For behavioral generation, a greedy response is marked correct if it contains any normalized gold alias, after lowercasing and removing punctuation, articles, and extra whitespace.

All main token-level analyses focus on the first content token of the gold answer under the model's tokenizer. This makes the final readout measurement well-defined: for the answer to begin correctly, the model must assign this token the highest next-token score at the answer position. Because this restriction does not by itself establish full-answer correctness, we report separate prefix, single-token, sequence-level, and continuation diagnostics.

## 4.1 Behavioral availability probes

We first compare direct generation with recognition-like behavioral probes. Direct accuracy is measured by greedy decoding from the canonical factual prompt. Paraphrase producibility is measured by evaluating a fixed set of paraphrased prompts for the same fact and marking the fact as producible if any paraphrase yields a normalized gold alias.

We also compute a teacher-forcing availability score by comparing the likelihood of the gold answer with the likelihood of matched alternatives under the same prompt. These behavioral probes are used to motivate the problem: they show that direct generation understates answer availability, but they do not define the main object of analysis.

## 4.2 Residual read and final write

For readout analyses, we run a single forward pass on the prompt and inspect the residual stream at the final prompt position, immediately before the model emits its answer. Let $h_i^{(\ell)}$ denote the residual-stream activation for item $i$ at layer $\ell$, and let $W_U$ be the unembedding matrix. At each layer, we apply the model's final normalization and unembedding:

$$z_i^{(\ell)} = W_U \mathrm{LN}_f(h_i^{(\ell)}). \tag{13}$$

Intermediate read is the strongest mid-layer logit-lens evidence for the gold answer token:

$$\mathrm{Read}_i^{\mathrm{int}} = \max_{\ell/L \in [0.4, 0.9]} \log \mathrm{softmax}(z_i^{(\ell)})_{a_i}. \tag{14}$$

We also compute final-layer read,

$$\mathrm{Read}_i^{\mathrm{fin}} = \log \mathrm{softmax}(z_i^{(L)})_{a_i}, \tag{15}$$

as a near-output reference.

Write measures whether the gold answer token is selected by the final readout. We operationalize it by the number of vocabulary tokens assigned a higher final logit:

$$\mathrm{Write}_i = |\{v : z_{i,v}^{(L)} > z_{i,a_i}^{(L)}\}|. \tag{16}$$

Thus $\mathrm{Write}_i = 0$ means that the gold answer token is top-ranked.

### 4.3 Type-controlled readability

A central concern is type priming. A birthplace prompt may make city names generally more decodable, even if the model has not recovered the correct city. To control for this, we compare the gold answer with type-matched decoys drawn from the same PopQA relation pool. For each item, the decoy set $\mathcal{D}_i^{\mathrm{type}}$ contains alternative answers to the same relation, excluding aliases of the gold answer.

The main priming margin compares the gold read score with the mean decoy read score:

$$m_i = \mathrm{Read}_i^{\mathrm{int}} - \frac{1}{|\mathcal{D}_i^{\mathrm{type}}|} \sum_{d \in \mathcal{D}_i^{\mathrm{type}}} \max_{\ell/L \in [0.4, 0.9]} \log \mathrm{softmax}(z_i^{(\ell)})_d. \tag{17}$$

An item is treated as fact-specifically readable when $m_i > 0$. As a stricter robustness check, we also require the gold answer to exceed the maximum same-type decoy rather than the mean decoy.

We also sweep read definitions to test whether the dissociation depends on a particular logit-lens choice. The sweep varies the intermediate layer band, whether layer scores are aggregated by maximum or mean, whether readability is defined against the mean or maximum decoy score, and whether decoys are drawn from all same-relation answers or frequency-matched same-relation answers. Throughout, intermediate read is interpreted as decodability under the logit lens, not as a claim that the model has already resolved the answer in a final form.

### 4.4 Prefix, single-token, and sequence-level checks

Because first-token analyses can be confounded by shared prefixes, we measure whether the first analyzed token uniquely identifies the gold answer among same-relation decoys, and also whether the first two tokens are unique. We define a stricter prefix-safe subset by excluding answers whose first token is short or prefix-ambiguous under these checks. We report read/write correlations on this subset and on the genuinely single-token subset for each tokenizer.

To test whether the phenomenon has a sequence-level analogue, we compute a sequence read score for the full gold answer string by summing or averaging teacher-forced log-probabilities of the gold answer tokens under the same intermediate readout. We compare this sequence read score with full greedy generation correctness. This sequence-level analysis is used as a robustness check; the causal projection interventions remain first-token interventions.

### 4.5 Selection margins and alternative candidates

For each item, we define the highest-ranked alternative candidate

$$b_i = \arg\max_{v \neq a_i} z_{i,v}^{(L)}, \tag{18}$$

and the gold answer's final selection margin

$$M_i(a_i) = z_{i,a_i}^{(L)} - z_{i,b_i}^{(L)}. \tag{19}$$

The gold answer is selected exactly when $M_i(a_i) > 0$. When $M_i(a_i) < 0$, the alternative candidate $b_i$ receives greater final-readout support.

For semantic analyses, we also identify the highest-ranked content alternative after excluding whitespace, punctuation, byte, and special tokens. Each selected alternative is classified by whether it is higher-frequency than the gold answer and whether it belongs to the same relation-specific answer pool.

### 4.6 Paired paraphrase controls

To test whether final support can vary for the same fact, we perform a paired paraphrase analysis. For each model, we restrict to facts for which at least one paraphrase produces the gold answer and at least one paraphrase does not. This holds fixed the subject, relation, gold answer token, answer frequency, and answer type.

For each success–failure pair, we measure intermediate read, answer-direction support, final selection margin, and the final support assigned to the failed prompt's highest-ranked alternative. We also analyze the subset in which the selected alternative is higher-frequency than the gold answer.

Finally, we run a within-fact cross-patch. For a failed paraphrase, we replace its answer-direction projection with the corresponding projection from a successful paraphrase of the same fact, while preserving the orthogonal component of the failed residual state. Recovery is measured by whether the gold answer becomes top-ranked after the patch.

### 4.7 Projection interventions

Let

$$h_i = \mathrm{LN}_f(h_i^{(L)}) \tag{20}$$

be the normalized final residual state, and let

$$u_{a_i} = \frac{W_U[a_i]}{\|W_U[a_i]\|} \tag{21}$$

be the unit unembedding direction of the gold answer token. The answer-direction support is

$$\pi_i(a_i) = \langle h_i, u_{a_i} \rangle. \tag{22}$$

For answer-up interventions, we replace this projection with a model-calibrated target $m_a$:

$$h_i' = h_i - \langle h_i, u_{a_i} \rangle u_{a_i} + m_a u_{a_i}. \tag{23}$$

The main targets are model-specific percentiles of answer-direction support among correctly generated examples: the 25th, 50th, and 75th percentiles, plus the mean. This frames the intervention as a calibrated counterfactual: would a failed item be selected correctly if it had answer-direction support typical of successful items from the same model?

For alternative-down interventions, we identify the highest-ranked content alternative $b_i$ and reduce the residual projection along its unembedding direction $u_{b_i}$. We use both a fixed downshift and a symmetric target based on the alternative-direction support distribution in successful generations. We also test a combined intervention that raises answer-direction support and reduces alternative-direction support. Matched random controls apply perturbations of the same norm in random directions orthogonal to the answer direction.

First-token recovery is the fraction of patched examples for which the gold answer token becomes top-ranked under the final readout. Separately, we greedily continue generation after the patch and measure full-answer alias correctness. We classify continuation outcomes as full correction, continuation drift after first-token recovery, alias/normalization failure, weak-prefix failure, or first-token non-recovery.

### 4.8 Layerwise trajectory analysis

For readable-but-unselected items, we track the layerwise difference between the highest-ranked alternative and the gold answer:

$$\Delta_i^{(\ell)} = z_{i,b_i}^{(\ell)} - z_{i,a_i}^{(\ell)}. \tag{24}$$

Items are grouped into trajectory categories. In alternative-dominant cases, the selected alternative has greater support than the answer across most of the late trajectory. In answer-downshifted cases, the answer has greater support at intermediate depth but falls below the alternative by the final readout. In late-crossing cases, the alternative exceeds the answer only near the final layers. Remaining items are assigned to an other category.

This analysis is descriptive: it characterizes how final selection emerges across layers, but does not by itself identify a causal circuit.

### 4.9 Output-frequency geometry

Token frequencies are estimated separately for each tokenizer from a fixed WikiText-103 sample with add-one smoothing. We measure norm-based frequency coupling by correlating unembedding row norm with token log-frequency:

$$\mathrm{corr}(\|W_U[w]\|, f_w). \tag{25}$$

Directional frequency coupling is measured by defining a frequency direction

$$r = \frac{\sum_w (f_w - \bar{f}) W_U[w]}{\left\|\sum_w (f_w - \bar{f}) W_U[w]\right\|}, \tag{26}$$

and correlating row projections $W_U[w] \cdot r$ with token log-frequency.

To test whether the frequency direction causally affects final readout selection, we remove the component of the final residual state along $r$ and decode from the patched state. We compare the rate at which the selected token changes, the frequency shift of the newly selected token, and gold-token recovery against matched random-direction ablations.

For checkpoint analyses, the frequency direction is computed once from a reference checkpoint and reused across training to avoid sign instability. At each OLMo checkpoint, we measure norm-frequency coupling, directional-frequency coupling, baseline-logit frequency coupling, effective rank, and spectral anisotropy. We apply the same norm and directional measurements to a controlled softmax toy model in which weight decay and co-evolving representations are toggled independently.

### 4.10 Statistical reporting

Rates and gaps are reported with bootstrap confidence intervals over items. Priming margins are tested against zero using a Wilcoxon signed-rank test. Comparisons that could be confounded by answer frequency are evaluated using frequency-bin matching with a within-bin label-permutation null. Cross-model claims are reported per model rather than only as pooled averages.

## 5 Results

### 5.1 Recognition probes motivate, but do not define, availability

Table 1 motivates the move from behavioral recognition to the read/write distinction. Direct greedy generation answers 27.0% of prompts correctly, while paraphrase sampling raises answer producibility to 44.3%. Thus, some facts are more available than direct generation alone suggests.

However, recognition-like probes do not define a stable itemwise class of facts that are "known but not generated." Paraphrase producibility and teacher-forcing advantage are only moderately aligned. We therefore use these probes as motivation, not as the main measurement target. The rest of the analysis asks a narrower question: when the answer is decodable inside the computation, does it receive enough final-readout support to be selected?

Table 1: Behavioral availability and read/write dissociation. Direct generation is greedy exact/alias matching. Readable failures are direct-generation failures with positive type-controlled priming margin. Write is the final rank of the gold answer token.

| Group | Quantity | Value | Extra |
|---|---|---|---|
| Behavioral | Direct greedy accuracy | 0.270 | $n = 600$ |
| Behavioral | Paraphrase-sampling producibility | 0.443 | $n = 600$ |
| Behavioral | Paraphrase gap over direct generation | 0.173 | CI [0.142, 0.205] |
| Behavioral | Paraphrase strength vs. TF advantage | 0.375 | correlation |
| Behavioral | Paraphrase-known vs. TF-advantage > 0 | 0.443 | agreement |
| Counts | Directly retrieved | 162 | |
| Counts | Behavioral gap items | 104 | |
| Counts | Absent under behavioral probe | 334 | |
| Counts | Direct-generation failures | 438 | |
| Counts | Readable direct-generation failures | 304 | $m_i > 0$ |
| Counts | Readable but not top-ranked | 294 | |
| Read/write | Mean priming margin | 2.925 | |
| Read/write | Median priming margin | 2.247 | |
| Read/write | Positive priming margin fraction | 0.760 | $n = 600$ |
| Read/write | Wilcoxon test vs. zero | $6.75\times10^{-54}$ | $p$-value |
| Read/write | Spearman(Read$^{int}$, −Write) | 0.737 | |
| Read/write | Spearman(Read$^{fin}$, −Write) | 0.978 | |
| Read/write | Not-top-ranked rate among readable failures | 0.967 | |
| Read/write | Median final answer rank | 62 | |

### 5.2 Many failed generations are readable but not selected

The read/write measurements reveal a large dissociation. Among the 438 direct-generation failures, 304 have positive type-controlled priming margin: the gold answer token is more decodable from intermediate residual states than same-relation decoys. These cases are not merely type-primed under the mean-decoy criterion; the gold token is more readable than alternatives drawn from the same answer pool.

Yet these readable answers usually remain below another token in the final next-token ranking. Of the readable direct-generation failures, 294 do not place the gold answer token at rank 1, giving a not-top-ranked rate of 96.7%. The median final answer rank is 62, so the answer is typically not just below the selected token by a negligible amount. It is detectable at intermediate depth but receives insufficient support at the final readout.

The correlation comparison in Table 1 gives the same picture without thresholding. Intermediate read is strongly related to final rank, but it is not equivalent to final selection. Final-layer read, by contrast, nearly restates the output distribution. This is the expected pattern if intermediate answer evidence and final-readout selection are coupled but separable.

### 5.3 The dissociation is robust to the read definition

The read/write dissociation is not an artifact of a single logit-lens measurement choice. Table 2 summarizes a sweep over 24 read definitions, varying the intermediate layer band, whether layer scores are aggregated by maximum or mean, whether readability is defined against the mean or maximum decoy score, and whether decoys are drawn from all same-relation answers or frequency-matched same-relation answers.

Across these variants, intermediate read remains less output-equivalent than final-layer read in nearly all model–variant pairs. The gap between final-layer and intermediate read correlations is positive across all variants for seven of eight models. The remaining case, Qwen2.5-3B, has a minimum gap of −0.002 under the most output-proximal read definition, effectively a tie. The readable fraction varies substantially across definitions, as expected, but the qualitative conclusion is stable: intermediate decodability is informative about final rank while remaining separable from final-readout selection.

Table 2: Robustness of the read/write dissociation across read definitions. For each model, we sweep 24 variants varying layer band, layer aggregation, decoy criterion, and decoy pool. The gap is $\rho(\mathrm{Read}^{fin}, -\mathrm{Write}) - \rho(\mathrm{Read}^{int}, -\mathrm{Write})$.

| Model | $\rho_{\mathrm{int}}$ range | Gap range | Readable fraction range |
|---|---|---|---|
| Llama-3.1-8B | 0.444–0.729 | 0.249–0.533 | 0.191–0.852 |
| Llama-3.2-3B | 0.477–0.779 | 0.198–0.499 | 0.220–0.832 |
| Llama-3.2-1B | 0.257–0.817 | 0.141–0.701 | 0.109–0.886 |
| Llama-3.2-3B-Inst. | 0.418–0.733 | 0.242–0.557 | 0.295–0.855 |
| Qwen2.5-3B | 0.358–0.745 | -0.002–0.385 | 0.119–0.798 |
| Qwen2.5-3B-Inst. | 0.277–0.743 | 0.112–0.578 | 0.122–0.822 |
| Qwen2.5-7B | 0.251–0.781 | 0.150–0.680 | 0.106–0.745 |
| Mistral-7B-v0.1 | 0.293–0.832 | 0.138–0.678 | 0.196–0.855 |

Table 3: Readable direct-generation failures under mean-decoy and hard-decoy criteria. The hard criterion requires the gold answer to exceed the maximum same-relation decoy.

| Model | $n$ | Mean decoy | Hard decoy |
|---|---|---|---|
| Llama-3.1-8B | 214 | 0.780 | 0.374 |
| Llama-3.2-3B | 228 | 0.781 | 0.425 |
| Llama-3.2-1B | 249 | 0.598 | 0.229 |
| Llama-3.2-3B-Inst. | 240 | 0.787 | 0.438 |
| Qwen2.5-3B | 241 | 0.734 | 0.282 |
| Qwen2.5-3B-Inst. | 246 | 0.740 | 0.325 |
| Qwen2.5-7B | 232 | 0.711 | 0.336 |
| Mistral-7B-v0.1 | 242 | 0.781 | 0.500 |

### 5.4 The effect survives stricter type controls

A central concern is that positive mean-decoy margin may be too permissive: the gold answer could exceed the average decoy while still falling below the strongest same-type decoy. Table 3 addresses this with a hard decoy criterion, requiring the gold answer to exceed the maximum same-relation decoy.

As expected, the hard criterion reduces the readable-failure rate. But it does not remove the phenomenon. Across models, the mean-decoy readable rate ranges from 59.8% to 78.7%, while the hard-decoy readable rate remains between 22.9% and 50.0%. Thus a substantial subset of failures remains fact-specifically readable even under a stricter comparison against the strongest same-type decoy.

Table 4 gives a complementary check. Readable-but-unselected items are not confined to the lowest priming-margin quartile. Even in the highest quartile, most readable direct-generation failures still do not put the gold answer at the top of the final distribution.

### 5.5 The same fact can be selected or not depending on final support

The paired paraphrase analysis provides a within-fact control. Table 5 restricts to facts for which one paraphrase elicits the correct answer and another paraphrase does not. This holds fixed the subject, relation, answer token, and answer frequency.

Across all eight models, successful paraphrases have stronger intermediate read than failed paraphrases of the same fact. They also have stronger final projection along the answer direction. In the failed paraphrases, the highest-ranked alternative has greater final-direction support than the answer across models. The failure is therefore not explained only by the fact being absent, the answer being rare, or the relation being difficult. The same fact can be selected or not depending on how the prompt routes evidence into the final answer direction.

Table 4: Robustness across priming-margin quartiles. The table reports the fraction of readable direct-generation failures whose gold answer is not top-ranked at the final readout.

| Priming-margin quartile | Not top-ranked | $n$ |
|---|---|---|
| [0.01, 1.19] | 1.000 | 76 |
| [1.19, 2.39] | 1.000 | 76 |
| [2.39, 4.36] | 0.987 | 76 |
| [4.36, 11.35] | 0.882 | 76 |

Table 5: Paired paraphrase analysis. We restrict to facts for which one paraphrase produces the gold answer and another paraphrase does not. This controls for fact, answer, frequency, and relation type.

| Model | $n$ | Read succ. | Read fail | Ans. supp. succ. | Ans. supp. fail | Alt. supp. fail | Margin succ. | Margin fail |
|---|---|---|---|---|---|---|---|---|
| Llama-3.1-8B | 60 | -5.23 | -7.17 | 17.20 | 13.66 | 16.71 | -1.17 | -3.98 |
| Llama-3.2-3B | 46 | -5.07 | -7.00 | 12.53 | 10.56 | 15.70 | -2.49 | -4.64 |
| Llama-3.2-1B | 41 | -6.77 | -7.82 | 18.67 | 17.28 | 21.80 | -1.89 | -3.55 |
| Llama-3.2-3B-Inst. | 37 | -3.65 | -4.13 | 12.64 | 11.63 | 15.14 | -1.32 | -2.70 |
| Qwen2.5-3B | 34 | -5.86 | -6.74 | 16.24 | 15.49 | 21.35 | -1.46 | -2.39 |
| Qwen2.5-3B-Inst. | 36 | -6.02 | -6.57 | 18.31 | 16.90 | 24.31 | -4.27 | -5.43 |
| Qwen2.5-7B | 32 | -7.27 | -7.42 | 23.29 | 22.77 | 26.57 | -0.40 | -0.91 |
| Mistral-7B-v0.1 | 59 | -5.30 | -5.61 | 45.60 | 40.39 | 57.46 | -1.29 | -2.78 |

The final-margin differences point in the same direction. Successful paraphrases have less unfavorable margins than failed paraphrases across models. The important quantity is not merely whether the answer is somewhat represented, but whether the final residual state gives it enough support relative to the highest-ranked alternative.

### 5.6 Readable failures are usually alternative-dominant

Table 6 classifies readable-but-unselected items by how the answer and the selected alternative evolve across the residual stream. The most common pattern is alternative dominance: the selected alternative already has greater support than the answer across much of the late trajectory. This category accounts for between 60 and 85 of 120 analyzed items per model.

By contrast, cases where the answer is clearly downshifted late are rare, ranging from 2 to 12 of 120 items per model. Late-crossing cases, where the selected alternative exceeds the answer only near the end, are rarer still. This suggests that many failures are not sudden last-layer reversals. More often, the computation enters the right semantic neighborhood while assigning stronger final-readout support to a plausible alternative.

### 5.7 Answer-direction support causally affects selection

Tables 7 and 8 test whether answer-direction support is causal. Retrieved, gap, and absent items differ substantially in residual projection along the answer direction, while their answer-token unembedding norms are similar. This supports the selection-margin account: what varies is mainly the final residual state's projection along the answer direction, not the fixed output-vector norm of the answer token.

The intervention results make this causal. Increasing answer-direction support raises gold-answer recovery far above the no-patch and random matched-norm controls in most models. Lowering support for the highest-ranked alternative alone recovers few answers, at most 4.5% across models. Combining answer-up and alternative-down is often more effective, recovering between 21.0% and 67.5% of cases depending on the model.

This pattern supports a relative-support account. The selected alternative matters, but many failures also require insufficient positive support for the correct answer. Targeted changes along meaningful output directions affect selection, while random matched-norm changes do not.

Table 6: Layerwise trajectory categories for readable-but-unselected items. Alternative-dominant cases have higher alternative support over much of the late trajectory; answer-downshifted cases have higher answer support earlier but not at the final readout.

| Model | $n$ | Alt.-dom. | Ans.-down | Late-cross | Other |
|---|---|---|---|---|---|
| Llama-3.1-8B | 120 | 68 | 9 | 0 | 43 |
| Llama-3.2-3B | 120 | 62 | 6 | 1 | 51 |
| Llama-3.2-1B | 120 | 72 | 5 | 0 | 43 |
| Llama-3.2-3B-Inst. | 120 | 62 | 9 | 1 | 48 |
| Qwen2.5-3B | 120 | 82 | 7 | 4 | 27 |
| Qwen2.5-3B-Inst. | 120 | 85 | 7 | 1 | 27 |
| Qwen2.5-7B | 120 | 70 | 2 | 1 | 47 |
| Mistral-7B-v0.1 | 120 | 60 | 12 | 0 | 48 |

Table 7: Answer-direction projection and aggregate patching. The decomposition separates residual projection along the answer direction from the fixed answer-token unembedding norm.

| Panel | Class or intervention | Projection / rate | Norm/logit |
|---|---|---|---|
| Decomposition | Retrieved | 13.885 | 1.106 / 15.246 |
| Decomposition | Behavioral gap | 10.342 | 1.128 / 11.512 |
| Decomposition | Absent | 9.468 | 1.134 / 10.599 |
| Patching | No patch: answer top-ranked | 0.050 | – |
| Patching | Answer-up: answer top-ranked | 0.344 | – |
| Patching | Random matched-norm: answer top-ranked | 0.044 | – |
| Patching | Alternative-up: alternative top-ranked | 0.300 | – |

### 5.8 Unselected answers are often ranked below plausible alternatives

Table 9 connects the mechanism to factual hallucination. In readable-but-unselected failures, the selected content token is frequently plausible: 60.9% are type-matched to the relation pool, and 68.0% are higher-frequency than the gold answer. The median selected-alternative–answer frequency gap is positive.

These failures therefore do not look like arbitrary noise. A birthplace prompt often leads to another city; an authorship prompt often leads to another author. The model is frequently in the right semantic neighborhood, but the correct instance is not assigned enough final-readout support to be selected.

Additional semantic-relatedness analyses are consistent with this interpretation but are less stable than the type and frequency results. External semantic similarity provides useful descriptive context, and it correlates with model-internal output geometry in several models, but candidate-selection regressions do not yield a stable model-agnostic semantic coefficient after controlling for frequency and output-geometry features. We therefore treat graded semantic similarity as a supporting descriptor rather than as the primary explanatory variable.

### 5.9 Frequency is represented directionally in the output map

Table 10 explains why frequency contributes to final selection without reducing the phenomenon to a simple norm effect. Across model families, token frequency is consistently represented by a shared output direction, while the correlation between frequency and unembedding row norm is weak and inconsistent. Equalizing output-vector norms also recovers very few answers.

This means frequent tokens are not favored only because their unembedding vectors are larger. They are favored because the output map contains a frequency-correlated direction that can add logit mass to common tokens. A frequent same-type alternative can therefore receive support both from relation-level compatibility with the prompt and from directional frequency structure.

Table 8: Per-model projection interventions. Answer-up increases answer-direction support; alternative-down lowers support for the highest-ranked content alternative; both applies both changes. Random controls use matched-norm directions.

| Model | $n$ | Answer-up | Alternative-down | Both | Random |
|---|---|---|---|---|---|
| Llama-3.1-8B | 200 | 0.660 | 0.030 | 0.600 | 0.010 |
| Llama-3.2-3B | 200 | 0.305 | 0.015 | 0.390 | 0.010 |
| Llama-3.2-1B | 200 | 0.355 | 0.000 | 0.315 | 0.030 |
| Llama-3.2-3B-Inst. | 200 | 0.715 | 0.025 | 0.675 | 0.020 |
| Qwen2.5-3B | 200 | 0.000 | 0.010 | 0.210 | 0.005 |
| Qwen2.5-3B-Inst. | 200 | 0.225 | 0.015 | 0.490 | 0.000 |
| Qwen2.5-7B | 200 | 0.635 | 0.015 | 0.550 | 0.015 |
| Mistral-7B-v0.1 | 200 | 0.315 | 0.045 | 0.395 | 0.010 |

Table 9: Properties of the selected content alternative when a readable answer is not top-ranked. Type-matched means that the selected alternative belongs to the same relation-specific answer pool as the gold answer.

| Property | Value | Extra |
|---|---|---|
| Readable-but-unselected items analyzed | 294 | |
| Selected alternative higher-frequency than answer | 0.680 | fraction |
| Selected alternative type-matched to relation pool | 0.609 | fraction |
| Median selected-alt.–answer frequency gap | +2.26 | log-units |

### 5.10 Frequency-direction ablation changes selection but rarely recovers the answer

The directional-frequency result does not imply that hallucinations are caused only by frequency-linked support. Table 11 tests this directly by ablating the residual component along the frequency direction and decoding from the patched state.

Frequency-direction ablation often changes the selected token, and the newly selected token is substantially lower-frequency than the original selected alternative. Matched random-direction ablations change the selected token less often and produce smaller or less systematic frequency shifts. However, gold-answer recovery remains low across models. Removing frequency-linked support changes the final ranking, but usually does not make the correct answer top-ranked.

This supports a mixed account. Frequency-linked geometry helps shape which candidate is selected, especially among plausible alternatives, but answer recovery also requires sufficient answer-direction support. Frequency is therefore a contributor to final-readout selection, not a standalone explanation of the read/write dissociation.

### 5.11 Training dynamics support a geometric account

Table 12 examines where this output geometry comes from. In OLMo checkpoints, directional frequency structure is already strong early in training and remains stable. Spectral concentration follows a related but separable trajectory, suggesting that the frequency direction is not merely a byproduct of global unembedding anisotropy.

The toy model isolates one possible mechanism for the decay of norm-based frequency encoding. Norm-carried frequency structure does not decay under weight decay alone or co-evolving representations alone. It decays when both are present. This supports the view that training can move frequency information away from simple vector norms and into directional structure in the output map.

Table 10: Output-frequency geometry across models. Norm–freq is the correlation between unembedding row norm and token log-frequency. Base–freq is the correlation between baseline logit and frequency. Dir.–freq is the frequency-correlated output direction. Norm-eq rec. is answer recovery after unembedding norm equalization.

| Model | Norm–freq | Base–freq | Dir.–freq | Norm-eq rec. |
|---|---|---|---|---|
| Llama-3.1-8B | 0.010 | 0.565 | 0.681 | 0.017 |
| Llama-3.2-3B | 0.212 | 0.563 | 0.723 | 0.005 |
| Llama-3.2-1B | 0.156 | 0.575 | 0.709 | 0.000 |
| Llama-3.2-3B-Inst. | 0.311 | 0.562 | 0.696 | 0.020 |
| Qwen2.5-3B | 0.086 | 0.595 | 0.509 | 0.000 |
| Qwen2.5-3B-Inst. | 0.085 | 0.498 | 0.512 | 0.005 |
| Qwen2.5-7B | -0.103 | 0.604 | 0.473 | 0.005 |
| Mistral-7B-v0.1 | -0.010 | 0.558 | 0.741 | 0.000 |

Table 11: Effect of ablating the frequency direction from the final residual state. Frequency-direction ablation often changes the selected token and shifts selection toward lower-frequency tokens, but rarely recovers the gold answer. Random controls use matched-norm directions.

| Model | Freq changed | Gold rec. | $\Delta f$ freq ablation | Rand changed | Rand rec. | $\Delta f$ random |
|---|---|---|---|---|---|---|
| Llama-3.1-8B | 0.715 | 0.020 | -3.935 | 0.275 | 0.000 | -0.842 |
| Llama-3.2-3B | 0.705 | 0.035 | -3.873 | 0.230 | 0.040 | -1.160 |
| Llama-3.2-1B | 0.870 | 0.015 | -4.724 | 0.435 | 0.025 | -1.893 |
| Llama-3.2-3B-Inst. | 0.670 | 0.020 | -3.089 | 0.160 | 0.025 | -0.278 |
| Qwen2.5-3B | 0.950 | 0.010 | -4.884 | 0.550 | 0.005 | 2.039 |
| Qwen2.5-3B-Inst. | 0.770 | 0.020 | -4.149 | 0.305 | 0.010 | 1.346 |
| Qwen2.5-7B | 0.975 | 0.005 | -5.092 | 0.525 | 0.015 | 1.594 |
| Mistral-7B-v0.1 | 0.625 | 0.060 | -3.424 | 0.340 | 0.015 | 1.143 |

### 5.12 Summary

Together, these results support a selection-margin account of one route to factual hallucination. Intermediate residual states can contain decodable, fact-specific evidence for the correct answer, but generation depends on whether that evidence receives enough support at the final readout. When it does not, the selected token is often a frequent or same-type alternative. The resulting error is plausible because the model has activated the right semantic neighborhood, but factually wrong because the correct instance is not assigned sufficient final-readout support.

## 6 Conclusion

We studied factual hallucination as a failure of output competition rather than as a binary absence of knowledge. Behavioral recognition probes suggest that some answers are more available than direct generation reveals, but they do not define a stable itemwise notion of what the model "knows." We therefore introduced a mechanical read/write distinction: read measures whether the gold answer token is decodable from intermediate residual states beyond type-matched decoys, while write measures whether that token wins the final next-token distribution.

Across models, many direct-generation failures are readable but not written. The correct answer can have fact-specific intermediate support and still lose the final competition to another token. Paired paraphrase controls show that this is not only an item-level property: the same fact can succeed or fail depending on how strongly a prompt routes evidence into the answer direction. Causal projection interventions identify this answer-direction write strength as a control variable for recovery.

These findings support a margin-based account of one route to factual hallucination. The model may enter the right semantic neighborhood, but fail to give the correct instance enough final output strength to

Table 12: Training dynamics and controlled toy results. OLMo checkpoint measurements use the sign-stable frequency direction. Toy settings independently toggle weight decay and co-evolving representations. Migration indicates decay of norm-based frequency coupling.

| Panel | Setting/step | Norm early | Norm late | Dir. early | Dir. late | Extra |
|---|---|---|---|---|---|---|
| OLMo | 1k | – | – | 0.765 | – | spectral aniso. 0.046; eff. rank 1953.5 |
| OLMo | 10k | – | – | 0.782 | – | spectral aniso. 0.182; eff. rank 1675.5 |
| OLMo | 100k | – | – | 0.778 | – | spectral aniso. 0.229; eff. rank 1578.6 |
| OLMo | 738k | – | – | 0.791 | – | spectral aniso. 0.218; eff. rank 1600.9 |
| Toy | Fixed rep., no decay | 0.642 | 0.703 | 0.402 | 0.507 | no migration |
| Toy | Fixed rep., decay | 0.642 | 0.718 | 0.399 | 0.507 | no migration |
| Toy | Co-evolve, no decay | 0.642 | 0.638 | 0.400 | 0.966 | no migration |
| Toy | Co-evolve, decay .005 | 0.642 | 0.417 | 0.400 | 0.634 | migration |
| Toy | Co-evolve, decay .01 | 0.642 | 0.375 | 0.400 | 0.538 | migration |

beat plausible competitors. Frequency further shapes this competition through directional structure in the unembedding, making common same-type alternatives especially dangerous.

Our results do not imply that all hallucinations have this form, or that intermediate decodability is the same as knowledge. Instead, they isolate a narrower mechanism: factual evidence can be internally decodable yet fail to control generation. Understanding hallucination therefore requires not only asking what information is present in a model, but how that information is amplified, suppressed, and selected at the point of output.

## A Appendix

You may include other additional sections here.